



AJKD Atlas of Renal Pathology: Postinfectious Glomerulonephritis

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Clinical and Pathologic Features

Postinfectious glomerulonephritis (GN) presents as acute nephritic syndrome with low complement C3 in serum, commonly following streptococcal throat or skin infection. Other infections may also give rise to postinfectious GN. Postinfectious GN most commonly occurs in children or young adults. The prognosis is excellent in children when the underlying infection resolves. Older adults, often diabetic or immunocompromised, may also develop postinfectious GN, more often from visceral sites of infection, and have a poor prognosis with frequent progression to end-stage kidney disease.

Light microscopy: Segmental or global endocapillary hypercellularity, with frequent neutrophils. Crescents may be present.

Immunofluorescence microscopy: Typically, polyclonal IgG and prominent C3 in chunky, irregular mesangial and capillary wall pattern (“garland” or “starry sky” appearance). Immunoglobulin A may be the dominant immunoglobulin deposited, particularly in older and diabetic patients.

Electron microscopy: Endocapillary hypercellularity, frequent neutrophils, occasional mesangial and small subendothelial deposits, and hump-type large subepithelial deposits without surrounding glomerular basement membrane reaction.

Etiology/Pathogenesis

Antigens of infectious agents activate the alternative complement pathway. Immune complexes may be circulating, or antigens can bind within the glomerulus with antibody and complement activation occurring in situ. Streptococcal pyrogenic exotoxin B has been proposed as a causative antigen in postinfectious GN following group A streptococcal infection.

Differential Diagnosis

Subepithelial humps are highly suggestive of postinfectious GN, but may also be seen in C3 GN. Other immune complex diseases may have endocapillary hypercellularity shown by light microscopy, but lack frequent neutrophils. Some processes have disease-specific immunofluorescence findings (eg, lupus nephritis), and do not show dominant C3 or frequent hump-type deposits.

Key Diagnostic Features

- Global endocapillary hypercellularity, with frequent neutrophils
- Dominant irregular capillary wall and mesangial C3, often with immunoglobulin G, and uncommonly immunoglobulin A
- Subepithelial hump-like deposits

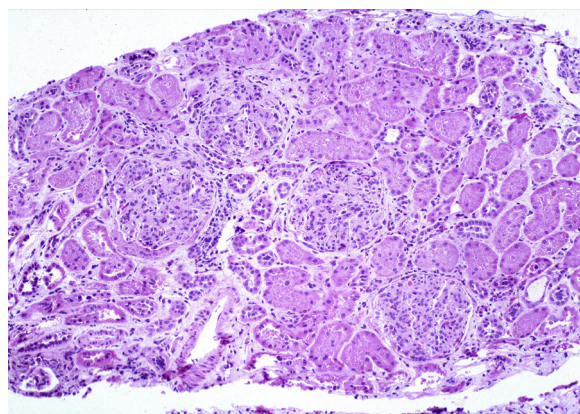


Figure 1. Postinfectious glomerulonephritis with diffuse mesangial and endocapillary hypercellularity, and a large number of polymorphonuclear neutrophils (hematoxylin and eosin stain). Reproduced with permission from *AJKD* 31(5):e1.

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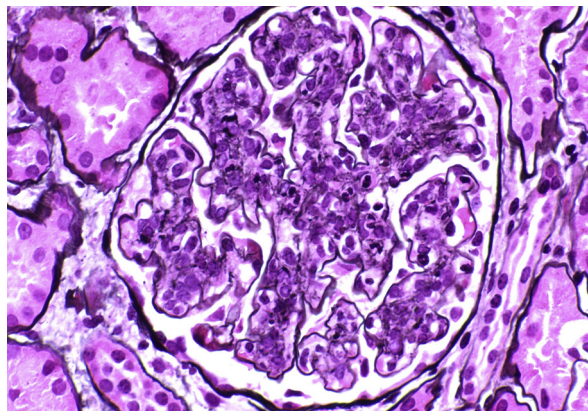


Figure 2. Postinfectious glomerulonephritis with endocapillary hypercellularity and many polymorphonuclear neutrophils (Jones silver stain). Reproduced with permission from *AJKD* 31(5):e1.

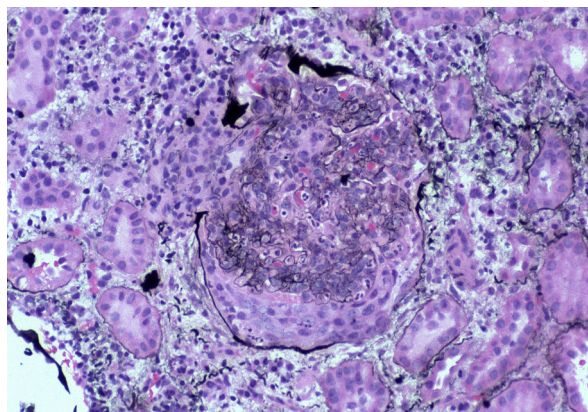


Figure 3. Crescents may occur in diffuse proliferative postinfectious glomerulonephritis. In this case, the crescent has ruptured the Bowman capsule, and there is an intense lymphoplasmacytic interstitial inflammatory response (Jones silver stain). Reproduced with permission from *AJKD* 31(5):e1.

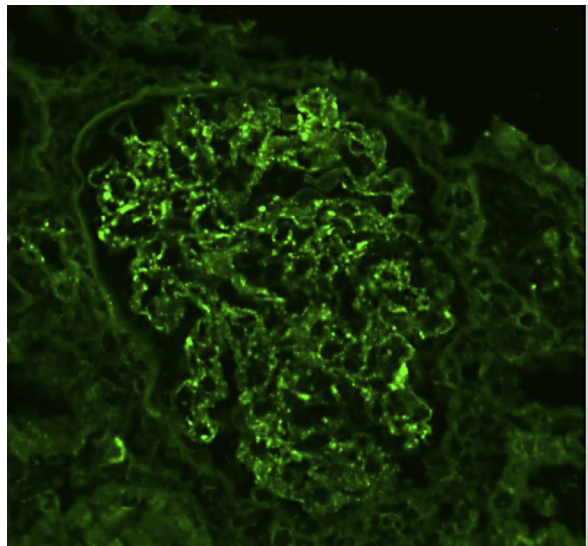


Figure 4. Postinfectious glomerulonephritis with irregular, coarse granular capillary loop and mesangial staining for immunoglobulin G (immunofluorescence microscopy).

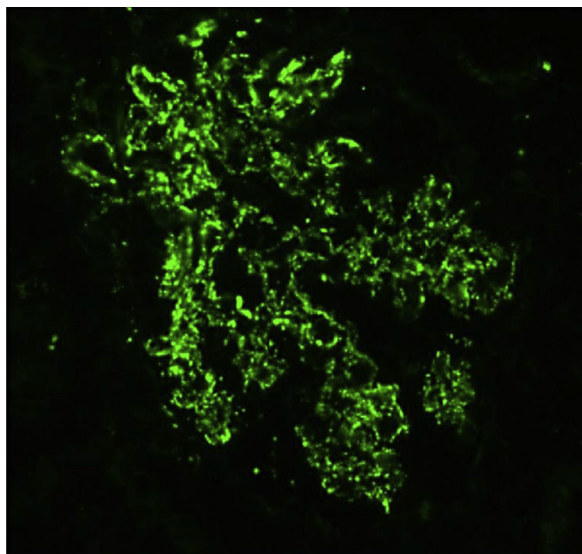


Figure 5. Scattered, irregularly spaced capillary wall and mesangial coarse granular staining (starry-sky pattern) for C3 in postinfectious glomerulonephritis (immunofluorescence microscopy).

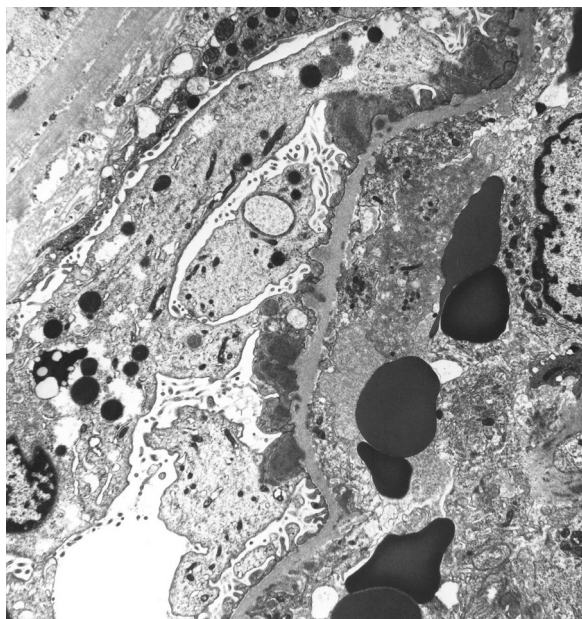


Figure 6. Subepithelial hump-like deposits in postinfectious glomerulonephritis. The deposits are irregularly spaced and frequently mottled. The deposits sit on top of the basement membrane with limited or no adjacent basement membrane reaction (electron microscopy). Reproduced with permission from *AJKD* 31(5):e1.